

# Module 1 | John the Ripper

|                    |                                                                  |
|--------------------|------------------------------------------------------------------|
| Student            | Mwansa Jonathan Musenge                                          |
| Module             | W3-PM1 - Kali Linux                                              |
| Target SHA-256     | 565843cf47c802ea5b7481d5e24c8e99a6a3f8a9eb861ef71bc0b1180be50e4e |
| Recovered password | good-luck                                                        |

## Objective and method

**Scope.** Recover the password of the instructor-provided PDF in **samples/module-1-course-pack-PDF1.pdf**. The brief permits John directly on Kali. The screenshots show a Kali VM; no use of the optional Johnny interface is evidenced.

1. The first `pdf2john.pl` command returned “command not found.” The student used `pdf2john` to extract a John-compatible record into `pdf_hash.txt`. The provided transcript includes pasted formatting artifacts; see `notes/original-terminal-transcript.txt`.
2. A first wordlist attack failed because `/usr/share/wordlists/rockyou.txt` was absent. The compressed `rockyou.txt.gz` was decompressed, then the attack was rerun.
3. John recognized one PDF hash (revision 4) and displayed **good-luck** after about 20 seconds. A subsequent `john --show --format=PDF pdf_hash.txt` reported one cracked hash and zero remaining. A supplied screenshot also shows a captured flag.

## Finding and recommendations

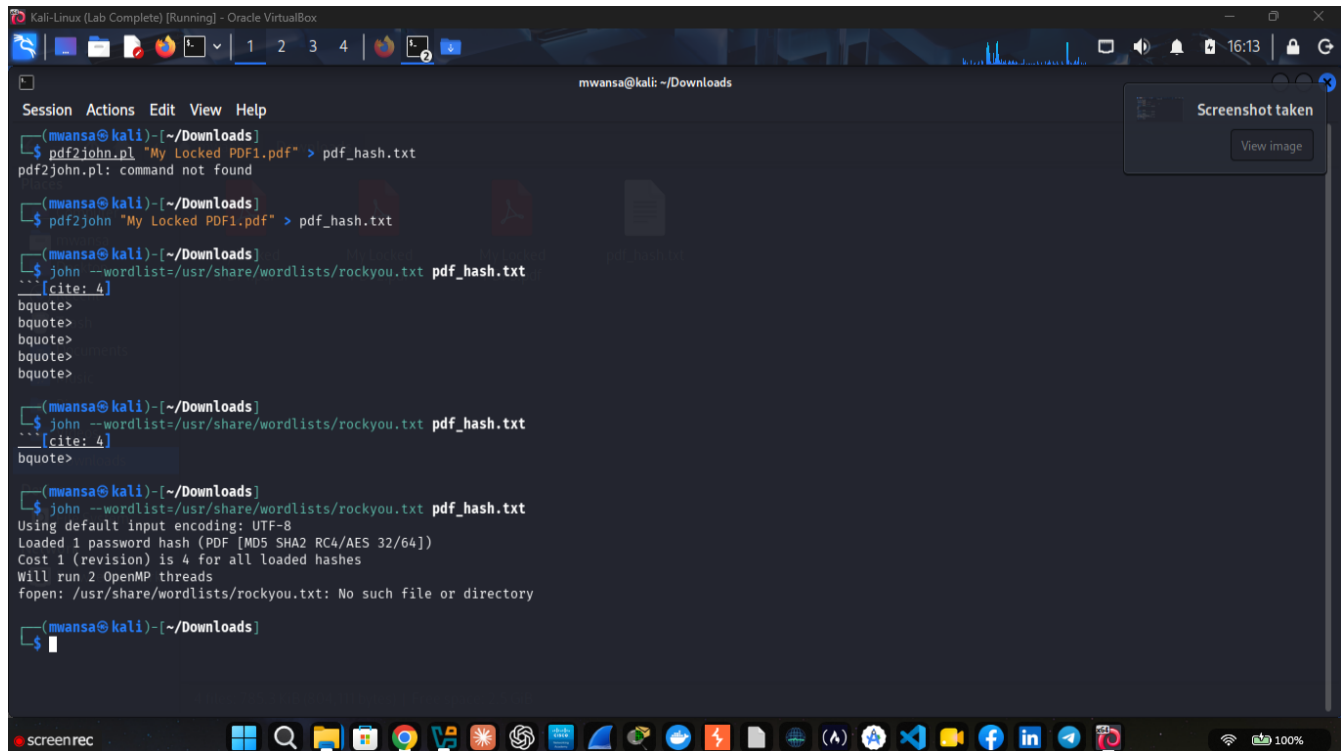
**F-01 - Guessable PDF password.** The course-pack sample fell to a common-word dictionary attack. For sensitive documents, use a unique long password generated outside common wordlists and restrict file access separately from password protection.

## Evidence limitations

The screenshots show the recovery and a flag page, but do not show the password being typed into a PDF viewer. The original extracted hash file and a complete live session log were not supplied. Timing is an observed tool result, not a benchmark. The other file named PDF1 is a different sample and belongs to Module 2.

# Evidence record

Screenshots supplied by the student; filenames and checksums are indexed in the repository.



```
(mwansa@kali)-[~/Downloads]
$ pdf2john.pl "My Locked PDF1.pdf" > pdf_hash.txt
pdf2john.pl: command not found

(mwansa@kali)-[~/Downloads]
$ pdf2john "My Locked PDF1.pdf" > pdf_hash.txt

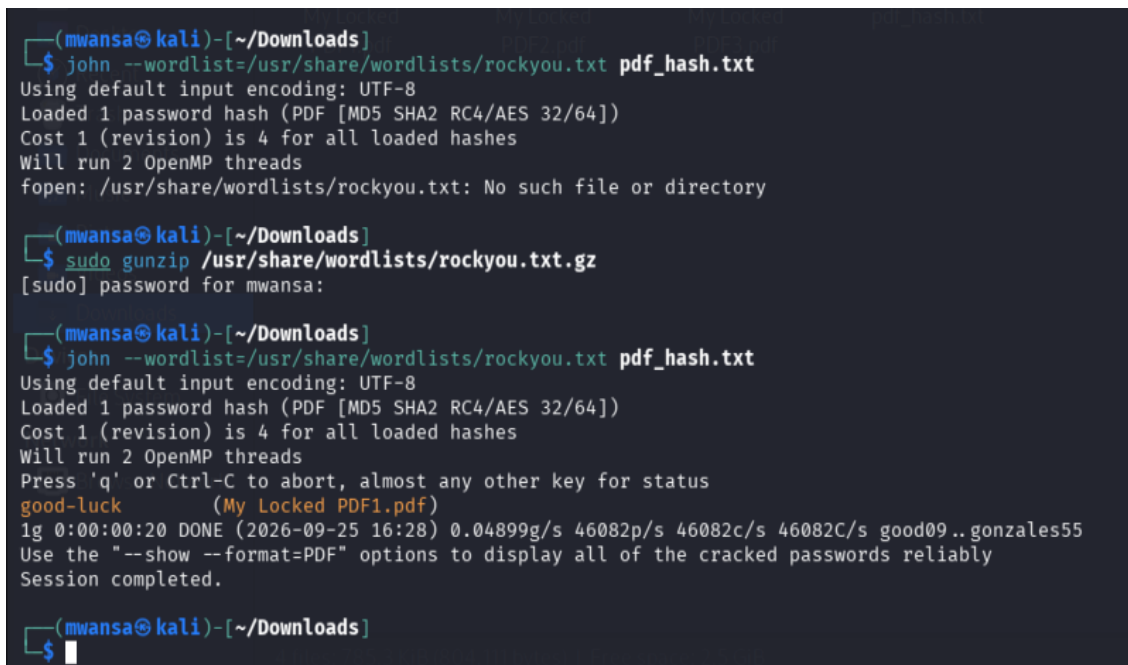
(mwansa@kali)-[~/Downloads]
$ john --wordlist=/usr/share/wordlists/rockyou.txt pdf_hash.txt
...[cite: 4]
bquote>
bquote>
bquote>
bquote>
bquote>

(mwansa@kali)-[~/Downloads]
$ john --wordlist=/usr/share/wordlists/rockyou.txt pdf_hash.txt
...[cite: 4]
bquote>

(mwansa@kali)-[~/Downloads]
$ john --wordlist=/usr/share/wordlists/rockyou.txt pdf_hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 2 OpenMP threads
fopen: /usr/share/wordlists/rockyou.txt: No such file or directory

(mwansa@kali)-[~/Downloads]
$
```

E02 | Initial wordlist path error in Kali.



```
(mwansa@kali)-[~/Downloads]
$ john --wordlist=/usr/share/wordlists/rockyou.txt pdf_hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 2 OpenMP threads
fopen: /usr/share/wordlists/rockyou.txt: No such file or directory

(mwansa@kali)-[~/Downloads]
$ sudo gunzip /usr/share/wordlists/rockyou.txt.gz
[sudo] password for mwansa:

(mwansa@kali)-[~/Downloads]
$ john --wordlist=/usr/share/wordlists/rockyou.txt pdf_hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
good-luck      (My Locked PDF1.pdf)
1g 0:00:00:20 DONE (2026-09-25 16:28) 0.04899g/s 46082p/s 46082c/s 46082C/s good09..gonzales55
Use the "--show --format=PDF" options to display all of the cracked passwords reliably
Session completed.

(mwansa@kali)-[~/Downloads]
$
```

E03 | John displays the recovered password.

```

(mwansa@kali)-[~/Downloads]
$ john --wordlist=/usr/share/wordlists/rockyou.txt pdf_hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 2 OpenMP threads
fopen: /usr/share/wordlists/rockyou.txt: No such file or directory

(mwansa@kali)-[~/Downloads]
$ sudo gunzip /usr/share/wordlists/rockyou.txt.gz
[sudo] password for mwansa:

(mwansa@kali)-[~/Downloads]
$ john --wordlist=/usr/share/wordlists/rockyou.txt pdf_hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (PDF [MD5 SHA2 RC4/AES 32/64])
Cost 1 (revision) is 4 for all loaded hashes
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
good-luck (My Locked PDF1.pdf)
1g 0:00:00:20 DONE (2026-09-25 16:28) 0.04899g/s 46082p/s 46082c/s 46082C/s good09..gonzales55
Use the "--show --format=PDF" options to display all of the cracked passwords reliably
Session completed.

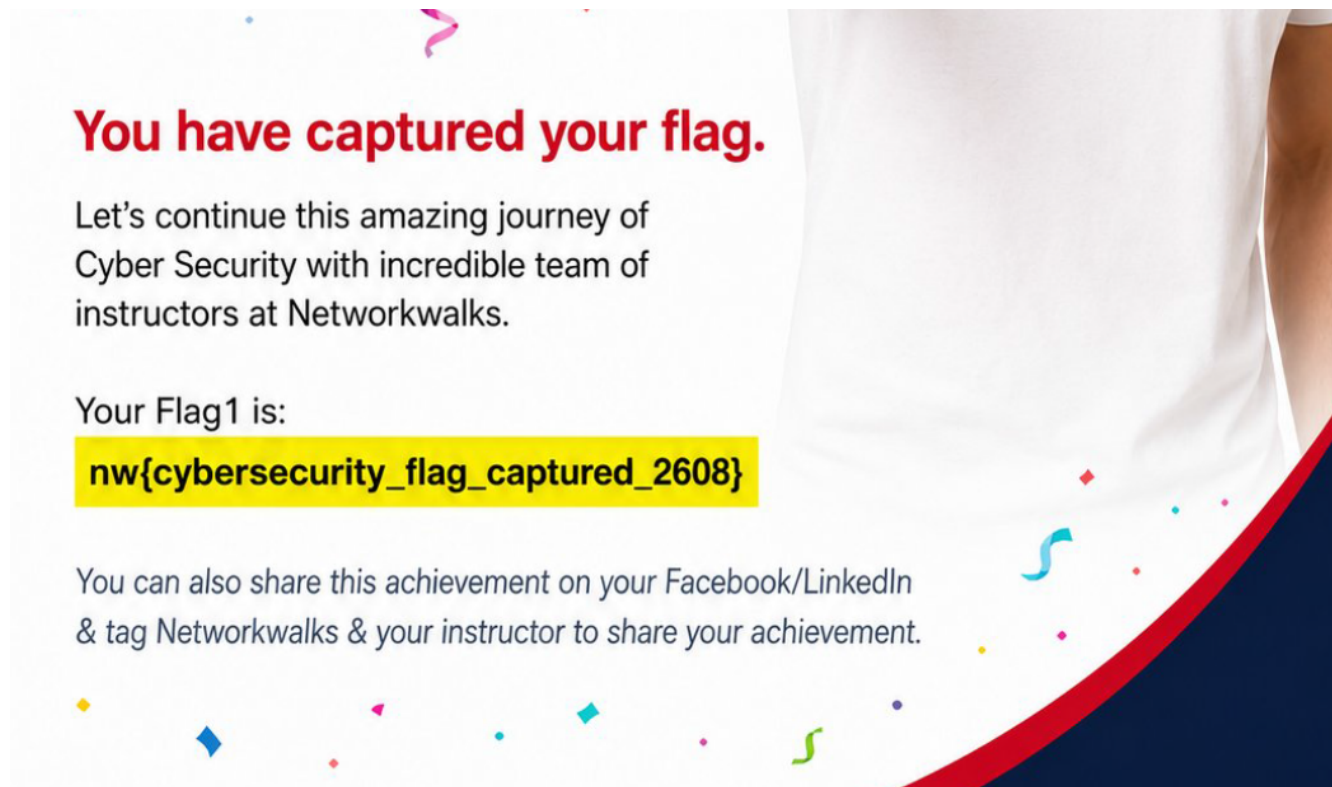
(mwansa@kali)-[~/Downloads]
$ john --show --format=PDF pdf_hash.txt
My Locked PDF1.pdf:good-luck

1 password hash cracked, 0 left

(mwansa@kali)-[~/Downloads]
$

```

E04 | John confirms one recovered hash; zero remain.



E05 | Supplied screenshot of the captured flag.